

- 2 Sponges are immobile, aquatic animals that live on rocks and sediment at the bottom of salt water environments.

Fig. 2.1 shows an example of a sponge, growing on the seabed, in a marine habitat.

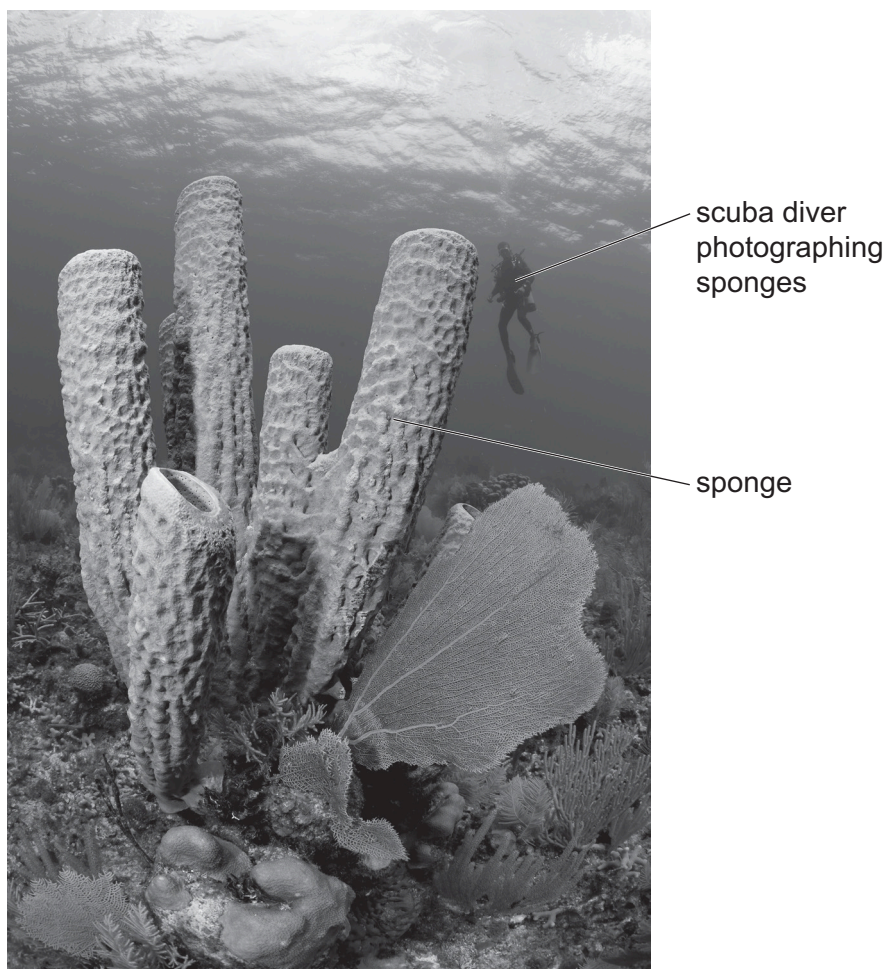


Fig. 2.1

There are many different species of sponge. Most species of sponge are sensitive to environmental stress. Human activity causes some environmental stress.

Scientists sampled the marine habitats along a length of coastline in Algeciras Bay in southern Spain to study the species diversity of sponges.

The scientists selected 12 sampling stations, **A** to **L**, as shown in Fig. 2.2. The scientists noted the land use or human activity along the coast next to each sampling station.

Key to diagram:

sampling station	type of land use or human activity
A	natural habitat
B	natural habitat
C	tourist beach
D	housing
E	shipping port
F	tourist beach
G	thermal power station
H	oil industry
I	ship building and repairs
J	tourist boats
K	tourist boats
L	natural habitat

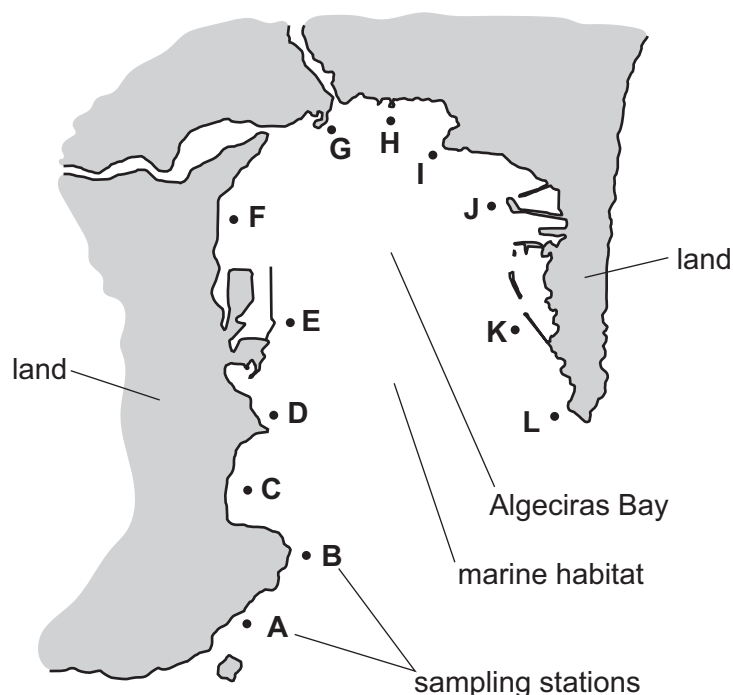


Fig. 2.2

The sampling stations were chosen to compare the effects of land use or human activity on the species diversity of sponges in the bay.

At each marine sampling station, the scientists:

- placed permanent line transects, 50 m in length, on the seabed
- photographed all sponges sighted at a distance of 1 m either side of the transect
- sampled each transect for the same length of time
- sampled each transect four times a year.

(a) The scientists standardised some variables.

State **two other** variables that the scientists should standardise in this investigation.

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- (b) The scientists compared the species diversity of sponges between different sampling stations using an index of diversity known as beta diversity.

The higher the beta diversity index, the higher the species diversity at a sampling station.

Fig. 2.3 shows a graph of the beta diversity index for each sampling station.

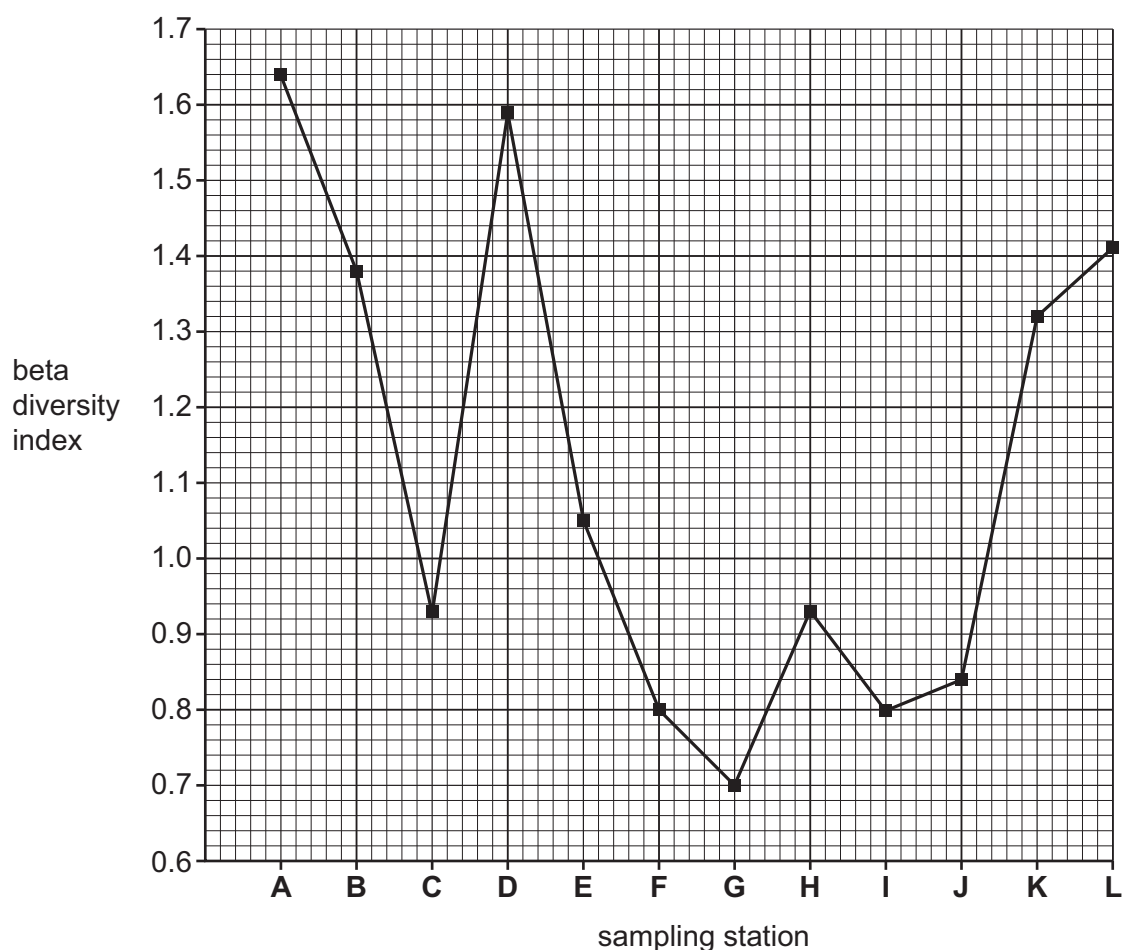


Fig. 2.3

A simplified formula for beta diversity is shown:

$$\text{beta diversity} = \frac{n}{\bar{x} - 1}$$

Key to symbols:

n = the number of sponge species recorded at a sampling station

$\bar{x}$ = the mean number of sponge species across all sampling stations

At sampling station **A**, 37 sponge species were recorded ($n = 37$).

Use Fig. 2.3 and the formula for beta diversity to calculate the mean number of sponge species across all sampling stations ($\bar{x}$).

Show your working. Write your answer to the nearest whole number.



- (c) (i) A student concluded that the species diversity of sponges was higher in the sampling stations where there was less human activity.

Using the information in Fig. 2.2 and in Fig. 2.3, evaluate this conclusion.

Use the values in Fig. 2.3 to support your answer.

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- (ii) Use the information in Fig. 2.2 and in Fig. 2.3 to state **two other** conclusions about the species diversity of sponges in Algeciras Bay.

Use the values in Fig. 2.3 to support your answer.

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[Total: 10]

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